MOUHAMADOU FADEL DIOP

Bioinformatics PhD Candidate | NGS Data Analysis | Pipeline Development



PROFESSIONAL SUMMARY

Bioinformatics PhD candidate with strong experience in next-generation sequencing (NGS) data analysis, pipeline development, and reproducible computational workflows. Skilled in processing and analysing genomic datasets and integrating multi-source biological and epidemiological data to derive meaningful insights. Proficient in R and Python, with experience working in Linux-based environments, HPC and handling large-scale sequencing data. Interested in applying computational approaches to multi-omics data integration and contributing to translational research in collaborative, interdisciplinary environments.

EDUCATION

MSc in Bioinformatics
Cheikh Anta Diop University, Dakar, Senegal — 2014

BSc in Bioinformatics-Biomathematics
Cheikh Anta Diop University, Dakar, Senegal — 2011

BSc in Physics and Chemistry
Cheikh Anta Diop University, Dakar, Senegal — 2010

RESEARCH EXPERIENCE

2022 | Present

PhD in Bioinformatics (Final Year)
London School of Hygiene and Tropical Medicine (LSHTM)
London, United Kingdoms

- Applied integrative approaches combining multiple data types to support biological and epidemiological insights

2019 | 2022

Bioinformatician
Medical Research Council Unit, The Gambia — Banjul, The Gambia

- Managed and processed large structured datasets, ensuring accuracy and consistency across multiple project files.
- Performed data cleaning, validation, and quality control checks on research and metadata records.
- Organized raw and processed data for reporting, analysis, and long-term digital storage.
- Prepared summaries, reports, and documentation to support project workflows and decision-making.
- Developed standardized workflows to improve data organization, reproducibility, and record traceability.
- Worked independently on detail-intensive data tasks while meeting deadlines.

CONTACT INFO

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CORE SKILLS

Next Generation Sequencing (NGS) Analysis
Bioinformatics Pipeline Development
Genomic Data Processing & Integration
Variant Analysis & Sequence Alignment
Data Cleaning & Reproducible Workflows
R (Bioconductor) & Python
Linux / HPC / Command Line
Data Visualisation & Reporting

2018 | 2019

Bioinformatician, intern

Medical Research Council Unit, The Gambia — Banjul, The Gambia

- Assisted with the organization, cleaning, and review of structured biological and metadata records.
- Processed and maintained tabular datasets used for analysis and reporting.
- Supported data quality checks to improve consistency and completeness across project files.
- Worked with spreadsheet-style data and research records in a structured digital environment.

PROFESSIONAL EXPERIENCE

2026

PhD Researcher – Bioinformatics

Infectious Disease Genomics & Transmission Dynamics

Banjul, The Gambia

- Developed reproducible computational pipelines for processing and analysing genomic sequencing data
- Performed quality control, sequence alignment, and variant-based analysis of whole-genome datasets
- Integrated genomic, temporal, and spatial data to infer biological relationships and transmission patterns
- Designed and implemented scalable data analysis workflows in R and Python
- Managed and organised large genomic datasets ensuring reproducibility and data integrity
- Generated analytical reports and visualisations to communicate complex results to interdisciplinary teams
- Collaborated with researchers across disciplines to support data-driven biological interpretation

2018 – 2022

Bioinformatician / Data Analyst

Medical Research Council at LSHTM — Banjul, The Gambia

- Managed and maintained structured datasets across multiple research and reporting projects.
- Reviewed, cleaned, and validated data to ensure completeness, consistency, and accuracy.
- Organized digital records, spreadsheets, and analysis-ready files for efficient access and long-term use.
- Generated summaries, tables, and reports from cleaned datasets to support team decision-making.
- Standardized data workflows and documentation to improve data quality and reduce errors.
- Collaborated with researchers and cross-functional teams to maintain high-quality project data.
- Worked extensively with Excel-compatible datasets, CSV files, tabular records, and structured metadata.
- Collaborated with team members to ensure data integrity across project activities.
- Worked independently on detail-intensive data tasks while meeting deadlines.

Technical Tools

R, Tidyverse, Bioconductor, Python

BWA, Minimap2, SAMtools, GATK, Qualimap

FASTQ, BAM, VCF processing

Git, GitHub, R Markdown

HPC, Linux / Unix environments

Languages

Wolof: Native

French: Proficient

English: Proficient

RESEARCH & DATA ANALYSIS PROJECTS

- Built genomic analysis workflows from raw sequencing data to downstream interpretation
- Applied statistical and computational methods to analyse biological datasets
- Developed reproducible pipelines using scripting and structured workflows
- Worked with structured and unstructured biological data in research settings

TECHNICAL EXPERTISE

Programming & Analysis - R (Bioconductor), Python
- Data analysis, visualisation
- Statistical & Mathematical modelling

NGS & Bioinformatics - Sequence alignment (BWA, Minimap2)
- Variant analysis workflows
- Genomic data formats (FASTQ, BAM, VCF)
- Quality control and preprocessing

Systems & Workflows - Linux / Unix environments
- Reproducible workflows (R Markdown)
- Version control (Git)

RESEARCH & COMMUNICATION

- Experience presenting research findings through reports and visualisations
- Strong ability to communicate complex computational results to non-specialists
- Experience working in collaborative, interdisciplinary research environments

ADDITIONAL INFORMATION

- Delivered beginner training on data handling and R-based data workflows.
- Supported colleagues and trainees in understanding structured data organization and reporting practices.
- Strong interest in multi-omics data integration and translational genomics
- Motivated to expand expertise in long-read sequencing, RNA-seq, and advanced genomics methods
- Committed to reproducible research and open science practices

Last updated on 2026-04-14.